Covariance Shrinkage and Minimum-Variance Portfolios: Evidence from the JSE

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1 Introduction

Historically, portfolio managers have relied on the sample covariance matrix to estimate the risks and diversification benefits associated with different assets. However, when the number of assets is large relative to the available return observations, the sample covariance matrix can contain substantial estimation error. Minimum-variance optimisation is particularly sensitive to this problem because it tends to exploit the most extreme covariance estimates, often producing highly concentrated and unstable portfolio weights. Consequently, portfolios that appear optimal in-sample may perform poorly out-of-sample.

Covariance-matrix shrinkage seeks to address this problem by combining the sample covariance matrix with a more stable, structured target. By shrinking extreme and unreliable estimates toward this target, the method can reduce estimation error and improve the robustness of portfolio allocation. Building on the work of Ledoit and Wolf, this essay explains the intuition and construction of shrinkage estimators, evaluates alternative approaches to reducing covariance-estimation error, and empirically investigates the performance of shrinkage-based portfolios using equities listed on the Johannesburg Stock Exchange. The empirical results show that linear shrinkage produced modest improvements in numerical stability, portfolio weights and out-of-sample performance, while nonlinear shrinkage remained close to the sample estimator.

2 The Sample Covariance Problem

The sample covariance matrix contains $N(N+1)/2$ distinct variances and covariances. Increasing the number of assets therefore raises the number of parameters at an approximate rate of N^2, without necessarily increasing the available historical observations. Because each estimate contains sampling error, these errors can accumulate as the matrix expands (Ledoit & Wolf, 2004a:378).

In mathematical terms, the sample covariance matrix, S, is an unbiased estimator of the true covariance matrix, Σ (Ledoit & Wolf, 2004a:368). Unbiasedness, however, does not imply precision. When the number of historical observations, T, is much larger than the number of assets, N, the sample covariance matrix, S, generally performs adequately.

Conversely, when the number of assets is large relative to the number of observations, the sample covariance matrix,

$$S = \frac{1}{T-1}\sum_{t=1}^{T}(r_t - \bar{r})(r_t - \bar{r})',$$

performs poorly as an estimator of the true covariance matrix (Ledoit & Wolf, 2004a:366). After the returns have been demeaned and centred around zero, the sample covariance matrix has a rank of at most $T-1$. Therefore, when $N \geq T$, its rank is smaller than its dimension, making the matrix singular and consequently non-invertible (Ledoit & Wolf, 2003:604). A third possible outcome occurs when $N < T$, but N remains close to T, such that the ratio N/T is large. The sample covariance matrix may then be invertible yet ill-conditioned, meaning that small changes in the estimated covariances can produce large changes in its inverse (Ledoit & Wolf, 2004a:366).

The accumulation of estimation errors across the matrix also distorts its eigenvalues. The eigenvalues represent the estimated variances of particular combinations of assets. In large-dimensional settings, the largest sample eigenvalues tend to be overestimated, while the smallest tend to be underestimated and pushed closer to zero (Ledoit & Wolf, 2004a:372). This can result in a drastic amplification of estimation errors when the matrix is inverted, indicating instability in the estimated covariance matrix (Ledoit & Wolf, 2003:614).

Investors use the inverse covariance matrix to solve the global minimum-variance problem and determine the optimal portfolio weights (Ledoit & Wolf, 2003:614):

$$\min_{w} w'Sw \quad \text{subject to} \quad \mathbf{1}'w = 1.$$

Its solution is

$$\widehat{w}_{\text{GMV}} = \frac{S^{-1}\mathbf{1}}{\mathbf{1}'S^{-1}\mathbf{1}}.$$

Because inversion replaces each eigenvalue with its reciprocal, an apparently low-risk combination associated with a very small eigenvalue can exert disproportionate influence over the weights. The optimiser cannot distinguish the true covariance structure from relationships created by sampling noise and may therefore assign large positive and negative weights to apparent diversification opportunities (Ledoit & Wolf, 2004b:111). If these relationships reflect noise, they may not persist out-of-sample, resulting in unstable weights and higher realised risk.

3 Shrinkage Estimation

3.1 The Shrinkage Principle

The sample covariance matrix is a natural data-driven estimator because it imposes little structure and remains unbiased when calculated using the $T-1$ denominator. However, it can have high estimation variance, which may translate into unstable portfolio weights and higher realised risk (Ledoit & Wolf, 2004b:111).

Shrinkage addresses this problem by forming a convex combination of a structured covariance target, F, and the sample covariance matrix, S. The target introduces bias because

its simplifying assumptions may not represent the true covariance structure, but it contains less sampling error and is more stable. The sample estimator is unbiased but can vary considerably between samples (Ledoit & Wolf, 2003:608). The shrinkage intensity, δ, controls the trade-off:

$$\widehat{\Sigma}_{\text{shrink}} = \delta F + (1-\delta)S, \qquad 0 \leq \delta \leq 1.$$

If $\delta = 0$, the estimator equals S; if $\delta = 1$, it equals F; and intermediate values combine the two.

Movement towards F introduces bias but reduces estimation variance. The relationship

$$\text{mean squared error} = \text{variance} + \text{squared bias}$$

shows why a biased estimator can outperform an unbiased one when its reduction in variance is sufficiently large (Ledoit & Wolf, 2004a:370). Estimation error is measured using the squared Frobenius distance,

$$L(\delta) = \|\delta F + (1-\delta)S - \Sigma\|_{\text{F}}^2 ,$$

which sums the squared differences between the shrinkage estimator and the true covariance matrix (Ledoit & Wolf, 2003:608–609). Because Σ is unobservable, the optimal intensity minimises the expected loss. More weight is assigned to F when S contains greater estimation error, while less shrinkage is required when S is more reliable (Ledoit & Wolf, 2004a:369). By reducing noisy extreme estimates, shrinkage produces a better-conditioned covariance estimate whose inverse is less likely to generate unstable weights (Ledoit & Wolf, 2004b:111; Ledoit & Wolf, 2003:614).

3.2 Linear Shrinkage and the Choice of Target

Several structured estimators can serve as shrinkage targets. One example is the scaled identity target. Ledoit and Wolf (2004a:368–370) shrink the sample covariance matrix towards mI, where m is the average variance or eigenvalue. In their idealised comparison, the target contributes bias but no estimation variance, whereas S is unbiased but contains estimation variance. The target assumes equal asset variances and zero covariance between different assets. It therefore improves conditioning but oversimplifies relationships such as the closer co-movement of banking shares relative to mining shares. The value of δ determines how strongly the sample estimate is pulled towards this target.

A K-factor model with uncorrelated residuals can also be used as a target. A multifactor model captures several sources of co-movement, but its result depends on which factors are selected. Ledoit and Wolf (2003:607–608) therefore use Sharpe's single-index model, under which stocks co-move through their different exposures to the market while company-specific residuals are assumed to be uncorrelated.

The constant-correlation target preserves individual sample variances but replaces the different pairwise correlations with their average. It recognises that equity returns generally co-move while imposing less structure than an unrestricted covariance matrix (Ledoit & Wolf, 2004b:113). The scaled identity target provides the strongest simplification, the single-index target retains market-driven co-movement and the constant-correlation target preserves a common level of correlation. The choice of target therefore balances stability against the risk of imposing an inaccurate covariance structure.

3.3 Nonlinear Shrinkage

Nonlinear shrinkage is rotation-equivariant: rotating the data rotates the estimate in the same manner. Equivalently, it retains the sample eigenvectors but adjusts the corresponding eigenvalues individually (Ledoit & Wolf, 2012:1025). This restriction is useful when no reliable external information is available. It also addresses the tendency for the largest sample eigenvalues to be overestimated and the smallest to be underestimated, exaggerating their dispersion.

When a multiple of the identity matrix is used as the target, linear shrinkage applies the same intensity, δ, to every eigenvalue, including those already close to their population values. Nonlinear shrinkage instead applies an eigenvalue-specific correction (Ledoit & Wolf, 2012:1025; Ledoit & Wolf, 2022:199, 205). An eigenvalue with greater estimated distortion may therefore receive a larger adjustment than one that is already well estimated. This flexibility can be valuable when population eigenvalues are dispersed and a common linear correction is too restrictive. Although the required corrections are more complex to estimate, nonlinear shrinkage can produce estimates closer to the population covariance matrix (Ledoit & Wolf, 2012:1025–1026).

4 Alternative and Complementary Approaches

Jagannathan and Ma (2003:1652–1653) distinguish between imposing structure on the covariance estimate, combining the sample matrix with a structured target and increasing the number of observations. This section considers factor models and principal components as structured estimators before examining portfolio constraints. The first two change the covariance estimator, whereas constraints act at the optimisation stage.

Factor models explain asset co-movement through a limited number of common factors and firm-specific shocks rather than estimating every pairwise relationship separately. Factors such as commodity prices, exchange rates or investor sentiment may cause shares with similar exposures to move together. For example, Fan et al. (2008:186–187) show that a three-factor model requires approximately $4N$ parameters rather than the $N(N+1)/2$ distinct parameters required by an unrestricted covariance matrix. The relevant factors must still be selected, and omitting an important factor may leave some co-movement unexplained. Factor estimators can therefore reduce sampling variance but introduce bias when their restrictions are inaccurate. Shrinkage limits this model risk by combining the structured estimator with the sample covariance matrix.

Principal-components analysis also reduces dimensionality but identifies common factors directly from patterns in the return data rather than selecting them beforehand. The first component explains the greatest amount of common variation, while subsequent components explain the greatest remaining variation. These components need not have clear economic interpretations. Using fewer components can reduce sampling noise, although discarded components may contain relevant covariance information. The number retained therefore remains a model-selection problem (Bai & Ng, 2002:191, 197–200). By contrast, nonlinear shrinkage preserves all sample eigenvectors and adjusts their eigenvalues continuously.

Portfolio constraints reduce the effect of estimation error by restricting the weights that the optimiser can select rather than changing the covariance estimator. Jagannathan

and Ma (2003:1674–1677) find that, after imposing non-negativity constraints, global minimum-variance portfolios using the sample covariance matrix perform almost as well as those using factor and shrinkage estimators. Their simulations show that this result continues to hold when returns are strongly influenced by a common factor. Constraints may therefore reduce the performance advantage of shrinkage, although they can also exclude useful positions and reduce the performance of estimators that are already stable.

The role of portfolio constraints also helps explain the empirical design below. By using unconstrained weights, the JSE application allows the effects of the different covariance estimators on short positions and portfolio concentration to remain visible.

5 Empirical Application: JSE Portfolios

5.1 Data and Design

Ledoit and Wolf's empirical portfolio evidence is drawn from US equities, so its applicability to the JSE cannot be assumed (Ledoit & Wolf, 2003). This illustration uses 17 manually selected large-cap JSE-listed companies spanning the financial, mining, telecommunications, retail and industrial sectors. It is not a complete or historical index sample. Daily adjusted closing prices from Yahoo Finance cover January 2015 to December 2024 and account for dividends and share splits.

Simple daily returns were divided chronologically: the first 70% (1,768 observations) formed the estimation window and the remaining 30% (758 observations) formed the out-of-sample evaluation window. Each covariance estimate generated an unconstrained global minimum-variance portfolio whose weights were held fixed throughout the test period. Holding the assets, dates and optimisation rule constant isolates the effect of changing the covariance estimator. Returns and volatility were annualised using 252 trading days.

5.2 Covariance Diagnostics

The condition number assesses whether a covariance matrix can be inverted reliably. It is the ratio of the largest eigenvalue to the smallest, with a larger value indicating that small estimation errors may cause larger changes in the inverse matrix and the resulting portfolio weights. Linear shrinkage reduced the condition number by approximately 14.5%, from 44.74 to 38.26, whereas nonlinear shrinkage produced a smaller reduction of approximately 3.9%, to 42.98.

Table 1 reports the covariance-matrix and portfolio diagnostics for the three estimators.

Table 1: Covariance-matrix and portfolio diagnostics

Estimator	Condition number	Minimum eigenvalue	Gross leverage	Minimum weight	Maximum weight
Sample covariance	44.74	0.0000795	1.241	-6.99%	28.48%
Linear shrinkage	38.26	0.0000911	1.205	-5.97%	27.19%
Nonlinear shrinkage	42.98	0.0000825	1.233	-6.87%	28.15%

Note: Portfolio weights are obtained from unconstrained global minimum-variance portfolios. Gross leverage is the sum of the absolute portfolio weights. Source: Author's calculations.

5.3 Portfolio Weights and Concentration

The improved conditioning under linear shrinkage also affected the portfolio weights. Because the optimisation was unconstrained, all three portfolios contained negative weights, representing short positions in which shares are borrowed and sold. The portfolios were also highly concentrated. Under the sample estimator, the largest short position was FirstRand at −6.99% and the largest positive position was Vodacom at 28.48%. Linear shrinkage made these weights less extreme at −5.97% and 27.19%, respectively. It reduced gross leverage, the sum of the absolute weights, from 1.241 to 1.205 and total short exposure from 12.07% to 10.25%. Vodacom and Mondi jointly represented 53.98% of the sample portfolio, compared with 51.97% under linear shrinkage. Linear shrinkage therefore moderated but did not eliminate concentration, while nonlinear shrinkage produced weights closer to the sample estimator.

5.4 Out-of-Sample Performance

The linear shrinkage estimator in this application used a scaled-identity target. To determine whether the differences in the covariance estimates and portfolio weights translated into improved performance, the portfolios were evaluated over 758 out-of-sample observations from 17 December 2021 to 31 December 2024. Table 2 reports the resulting performance measures.

Table 2: Out-of-sample portfolio performance

Estimator	Annualised return	Annualised volatility	Maximum drawdown	Sharpe ratio
Sample covariance	4.41%	16.47%	−16.92%	0.268
Linear shrinkage	4.71%	16.39%	−16.75%	0.287
Nonlinear shrinkage	4.47%	16.46%	−16.85%	0.272

Note: Returns and volatility are annualised using 252 trading days. The Sharpe ratio assumes a zero risk-free rate. Source: Author's calculations.

Because these were global minimum-variance portfolios, realised volatility was the primary outcome. Linear shrinkage produced the lowest annualised volatility, reducing it from 16.47% under the sample estimator to 16.39%. Nonlinear shrinkage produced annualised volatility of 16.46%, remaining close to the sample estimator. The improvement under linear shrinkage was therefore only 0.08 percentage points. Linear shrinkage also recorded the highest realised return of 4.71%, the least severe maximum drawdown of −16.75% and the highest zero-risk-free-rate Sharpe ratio of 0.287. These results ranked linear shrinkage first across the reported measures, although the differences were small. Its higher realised return should not be interpreted as an expected-return forecast because shrinkage altered only the covariance estimate and resulting portfolio weights.

Figure 1 presents the cumulative returns over the test period. The substantial overlap between the three paths visually reinforces the small economic differences reported in Table 2.

5.5 Interpretation and Limitations

Linear shrinkage produced the most stable covariance matrix, less extreme weights and the strongest out-of-sample results, although the improvements were marginal. Nonlinear shrinkage remained close to the sample estimator. This is consistent with the

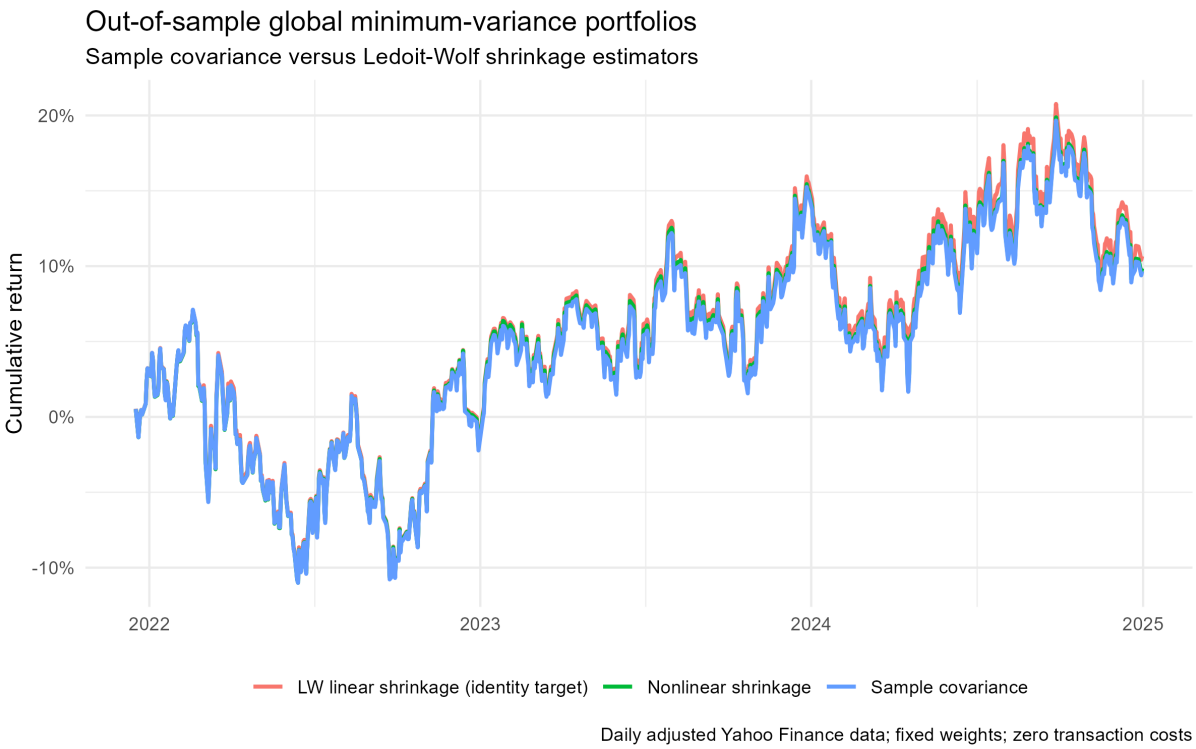


Figure 1: Cumulative out-of-sample returns of the minimum-variance portfolios

low-dimensional design: 17 assets were estimated using 1,768 observations, giving an N/T ratio of approximately 0.0096. The sample matrix was therefore already reasonably stable, leaving limited estimation error for shrinkage to correct. The manually selected sample, single test window and absence of statistical-significance testing limit the findings, as do unconstrained short selling, fixed weights and zero transaction costs. The results should therefore be interpreted as illustrative rather than conclusive.

6 Conclusion

Sample covariance matrices can produce unstable minimum-variance portfolio weights. This essay investigated whether shrinkage provides a more reliable alternative by using shrinkage targets and intensities to reduce sampling error. The principle of shrinkage is to trade some bias for lower estimation variance. Linear shrinkage applies the same intensity to all eigenvalues, whereas nonlinear shrinkage applies different adjustments depending on their estimated distortion. To examine whether the findings of Ledoit and Wolf extend to the South African market, the empirical application compared the sample covariance estimator, linear shrinkage towards a scaled-identity target and nonlinear shrinkage using selected JSE-listed equities. Linear shrinkage produced the most stable covariance matrix, slightly less extreme weights and the strongest out-of-sample performance, although its improvements were marginal. These small differences are consistent with using only 17 assets relative to 1,768 estimation observations, giving an N/T ratio of approximately 0.0096. The results suggest that shrinkage can be a useful risk-management tool, particularly in higher-dimensional settings, but does not guarantee superior portfolio performance.

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